Customer Engagement & Product Utilization Analytics for Retention Strategy

Internship Project Report

Submitted to: Unified Mentor

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Executive Summary:

1. Purpose

This project analyzes customer churn and retention patterns to identify customer characteristics and segments associated with higher observed churn and to develop data-driven insights that can support customer retention strategies.

2. Key Overall Finding

The analysis covered 10,000 customers, of which 2,037 customers had exited, resulting in an overall observed churn rate of 20.37%.

3. Major Findings

The analysis identified several important patterns:

- **Germany:** Germany recorded the highest observed churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain.
- Inactive customers: observed churn rate of 26.85%, compared with 14.27% among active customers.
- **High-balance inactive customers:** 2,903 customers were identified in this segment, with an observed churn rate of 31.48%.
- **Product utilization:** Customers with 2 products had the lowest observed churn rate at 7.58%, while customers with 3 and 4 products showed much higher observed churn rates of 82.71% and 100%, respectively. The 3- and 4-product groups were relatively small, so these results should be interpreted cautiously**.**
- **High-risk segment:** Customers in Germany with a Relationship Score of 0 and age above 40 showed an observed churn rate of 75.36% across 345 customers. This was substantially higher than the 30.67% observed churn among customers aged 40 or below within the same Germany + Score 0 segment.

4. Business Recommendations

Prioritize high-risk customer segments, particularly the Germany + Relationship Score 0 + age above 40 group, and investigate the factors associated with their high observed churn. The bank should also investigate high-balance inactive customers, learn from the lower churn observed among two-product customers, and improve meaningful engagement among inactive customers through customer feedback and additional behavioral data. Retention strategies should be targeted to customer needs and monitored using relevant engagement and churn measures.

5. Overall Business Value

The analysis helps the bank identify where retention problems are concentrated, while customer feedback and additional behavioral information can help determine why these problems occur. This can support more targeted, customer-focused retention strategies rather than applying the same approach to all customers.

1. Introduction:

Customer retention is an important concern for banks because losing customers can affect long-term customer relationships and business performance. Understanding customer churn can help organizations identify patterns associated with customers leaving and highlight areas where further investigation may be needed.

This project focuses on analyzing customer churn and retention using customer characteristics, engagement, and product-utilization data. Several variables, including geography, age, balance, number of products, customer activity, and credit-card ownership, were examined to identify patterns associated with churn.

The analysis aims to identify customer segments with relatively higher observed churn and translate these findings into meaningful business insights for retention strategy. However, the available dataset does not contain the specific reasons why individual customers left. Therefore, the analysis identifies **associations and potential risk segments rather than causal explanations**, and additional customer feedback or information would be required to understand the underlying reasons for churn.

2. Problem Statement:

The primary problem addressed in this project is to understand the extent of customer churn and identify customer characteristics and segments that show stronger associations with observed churn. The analysis examines variables such as geography, customer activity, age, balance, and number of products to identify differences in churn rates across customer groups.

The analysis also considers the size of each customer group when interpreting churn rates. For example, customers with three and four products showed very high observed churn rates of 82.71% and 100%, respectively, but these groups were substantially smaller than the one- and two-product groups. Therefore, the results require careful interpretation rather than assuming that higher product counts directly cause churn.

Further segmentation was used to investigate high-risk groups. This analysis identified customers in Germany with a Relationship Score of 0 and age above 40 as a particularly

high-risk segment, with an observed churn rate of 75.36%. The project therefore aims not only to measure churn but also to identify meaningful customer segments that can help the bank investigate potential retention opportunities.

3. Project Objectives:

1. To measure the overall customer churn rate.
2. To identify customer characteristics associated with customer churn.
3. To identify high-risk customer segments that may require further investigation.
4. To provide data-driven insights and recommendations to support customer retention strategies.
5. To develop an interactive dashboard to present key findings and support data-driven decision-making.

4. Dataset Description:

The dataset used in this project is the **European Bank** dataset. It contains **10,000 customer records and 14 variables**, providing information about customer demographics, financial characteristics, engagement, and banking product usage.

The dataset includes several important variables used for analyzing customer churn and retention. Geography identifies the customer's country and contains three categories: **France, Germany, and Spain**. Age represents the customer's age in years, while Tenure represents the number of years the customer has been associated with the bank. Balance represents the amount of money held in the customer's bank account, and EstimatedSalary represents the customer's estimated salary or income.

The dataset also contains variables related to customer engagement and product usage. IsActiveMember indicates whether a customer is active (1) or inactive (0). NumOfProducts represents the number of banking products held by a customer. HasCrCard indicates whether the customer has a credit card (1) or does not have one (0). CreditScore represents a customer's creditworthiness.

The target variable for this project is Exited. A value of 1 indicates that the customer **churned**, while a value of 0 indicates that the customer **stayed with the bank**. These variables were used to examine patterns and differences in customer churn across different customer groups.

Variable	Description
Geography	Customer's country: France, Germany, or Spain
Age	Age of the customer in years
Tenure	Number of years the customer has been with the bank
Balance	Amount of money held in the customer's bank account
EstimatedSalary	Estimated salary or income of the customer
CreditScore	Numerical measure of the customer's creditworthiness
NumOfProducts	Number of banking products held by the customer
HasCrCard	1 = has a credit card, 0 = does not have one
IsActiveMember	1 = active customer, 0 = inactive customer
Exited	1 = churned, 0 = stayed

5. Methodology:

1. **Data Loading and Preparation**
   - Dataset loaded using Python and Pandas.
2. **Exploratory Data Analysis**
   - Examined customer characteristics and churn patterns.
3. **Group Comparison**
   - Used groupby() and pd.crosstab().
4. **Percentage and Distribution Analysis**
   - Compared percentages and used histograms for numerical distributions.
5. **Customer Segmentation**
   - Used filtering to identify specific high-risk groups.
6. **Relationship Strength Analysis**
   - Created a simple descriptive Relationship Score using activity and product depth.
7. **Business Insight Development**
   - Converted findings into potential areas for investigation and retention opportunities.
8. **Interactive Dashboard**
   - Presented key KPIs and high-risk segments through Streamlit.

6. Results & Findings:

6.1 Overall Customer Churn

The analysis was conducted on 10,000 customer records, of which 7,963 customers remained with the bank and 2,037 customers had exited. The overall observed customer churn rate was 20.37%.

This indicates that approximately one in five customers in the dataset had churned. The overall churn rate provides a baseline for evaluating differences in churn across customer characteristics and segments.

6.2 Geography and Customer Churn

A comparison of observed churn rates across countries showed substantial differences. Germany recorded the highest observed churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain.

Germany's observed churn rate was approximately twice that of France and Spain. This finding suggests an association between geography and customer churn within the dataset. However, geography itself cannot be considered a direct cause of churn based on this analysis.

The substantially higher churn rate in Germany therefore became an important area for further investigation.

6.3 Investigation of Age Across Geographies

Since Germany showed substantially higher churn, customer age was examined as one possible differentiating factor.

The average age was:

- **France: 38.51 years**
- **Germany: 39.77 years**
- **Spain: 38.89 years**

Although Germany had the highest average age, the difference between Germany and France was only 1.26 years. Further examination of the age distributions also did not reveal a sufficiently large difference to explain Germany's substantially higher churn rate.

Therefore, average age alone did not appear sufficient to explain the difference in churn across countries, and further segmentation was required.

6.4 Customer Activity and Churn

Customer activity showed a more noticeable difference in observed churn. Active customers had an observed churn rate of 14.27%, compared with 26.85% among inactive customers.

This represents a difference of approximately 12.58 percentage points.

The finding suggests an association between customer engagement and retention. Inactive customers may therefore represent an important group for further investigation. The bank could collect feedback from both active and inactive customers to identify potential service gaps, engagement issues, or other factors that may be associated with inactivity.

However, the dataset does not contain information about the reasons for customer inactivity or churn, so no causal conclusion can be made.

6.5 High-Balance Customer Disengagement

The analysis also examined customers whose account balance was above the overall average balance of approximately 76,485.89.

A total of 2,903 customers were identified as having a high balance while also being classified as inactive. This group had an observed churn rate of 31.48%.

This finding indicates that a relatively high account balance does not necessarily correspond to strong customer engagement or retention. High-balance inactive customers may therefore represent an important segment for further investigation.

The bank could use customer feedback and additional behavioral information to understand why customers with relatively high balances remain inactive and whether targeted engagement or retention strategies would be appropriate.

6.6 Product Utilization and Churn

The relationship between the number of banking products held by customers and observed churn showed a non-linear pattern.

Number of Products	Customers	Observed Churn Rate
1	5,084	27.71%
2	4,590	7.58%
3	266	82.71%

Number of Products	Customers	Observed Churn Rate
4	60	100.00%

Customers with two products had the lowest observed churn rate at 7.58%. In contrast, the three- and four-product groups showed substantially higher observed churn.

However, the three- and four-product groups were relatively small compared with the one- and two-product groups. Therefore, these results require cautious interpretation and should not be interpreted as evidence that having more products causes churn.

The relatively low churn observed among two-product customers could be investigated further to understand whether particular services, combinations of products, or customer experiences are associated with stronger retention. This could provide useful information before developing product-bundling or cross-selling strategies.

6.7 Relationship Strength Analysis

To further examine the combined effect of customer activity and product depth, a simple Relationship Score was developed for this analysis.

The score combined:

- **Customer activity**
- **Number of products held**

Customers were grouped into three relationship-strength categories:

Relationship Score	Customer Profile	Observed Churn Rate
0	Inactive + 1 product	36.65%
1	Active + 1 product OR Inactive + 2+ products	17.64%
2	Active + 2+ products	9.66%

The results showed a clear difference in observed churn across the three groups. Customers with a Relationship Score of 0 had the highest observed churn rate at 36.65%, while customers with a Score of 2 had the lowest at 9.66%.

This suggests that stronger engagement combined with greater product depth is associated with lower observed churn in the dataset.

However, the Relationship Score is a descriptive measure created for this analysis and has not been statistically validated. Therefore, it should be considered an exploratory indicator rather than a definitive measure of customer relationship strength.

6.8 Identification of a High-Risk Customer Segment

Further segmentation was performed to identify a potentially high-risk customer group.

The analysis identified customers who met all three conditions:

- **Geography: Germany**
- **Relationship Score: 0**
- **Age: Above 40 years**

This segment contained 345 customers and had an observed churn rate of 75.36%.

For comparison, Germany-based customers in the same Relationship Score 0 group who were aged 40 or below had an observed churn rate of 30.67%.

The difference of approximately 44.69 percentage points represents a substantial difference in observed churn within this specific segment. This suggests a strong association between age and observed churn within the Germany + Relationship Score 0 segment.

The finding does not establish that age causes churn. Instead, it identifies a potentially important customer segment that warrants further investigation.

Business implication

The bank should first investigate why this segment shows such a high observed churn rate. Customer feedback and additional behavioral information could help identify the factors associated with churn among customers above 40 years of age in this segment.

Based on the findings and customer feedback, the bank could then adapt its retention strategies to address the specific needs or concerns of these customers.

6.9 Overall Findings

Overall, the analysis identified several customer characteristics and segments associated with differences in observed churn:

1. **Overall churn: 20.37% of customers had exited.**
2. **Geography: Germany had the highest observed churn rate at 32.44%.**
3. **Customer activity: Inactive customers had a higher observed churn rate than active customers.**

4. **High-balance disengagement: 2,903 high-balance customers were inactive, with an observed churn rate of 31.48%.**
5. **Product utilization: Customers with two products showed the lowest observed churn rate at 7.58%, while the three- and four-product groups showed very high churn but were relatively small.**
6. **Relationship strength: Customers with a Relationship Score of 0 had substantially higher observed churn than customers with higher scores.**
7. **High-risk segment: Germany-based customers with a Relationship Score of 0 and age above 40 had an observed churn rate of 75.36%.**

7. Limitations of the Analysis:

Although the analysis identified several important patterns associated with customer churn, there are certain limitations that should be considered when interpreting the findings.

a. **Lack of information about the reasons for churn:**
The dataset does not contain information about the specific reasons why customers left the bank. Therefore, the analysis can identify customer characteristics and segments associated with higher observed churn, but it cannot determine the underlying causes of churn. Customer feedback and additional behavioral information would be required to investigate the reasons behind these patterns.

b. **Small size of some customer segments:**
Some groups contained relatively few customers. For example, customers with three products had an observed churn rate of **82.71%**, while customers with four products had an observed churn rate of **100%**. However, these groups contained only **266 and 60 customers**, respectively. Therefore, these results should be interpreted cautiously and should not be treated as evidence that having more products causes customer churn.

c. **No predefined high-risk customer segments:**
The dataset did not provide predefined high-risk customer segments. Therefore, meaningful segments had to be identified through exploratory analysis of the available variables. While this allowed deeper investigation, it also required additional analysis time. In a real-world business environment, extensive exploratory analysis may delay the identification of priority areas and the implementation of retention strategies.

d. **Relationship Score was a descriptive measure created for this analysis:** The Relationship Score used in this project was developed as a simple descriptive measure based on customer activity and product depth. The scoring system was not an officially defined or statistically validated measure. Therefore, it should be considered an exploratory indicator rather than a definitive measure of customer relationship strength. Further validation would be required before using such a score for operational decision-making.

Overall limitation:

These limitations mean that the findings should primarily be used to **identify patterns, prioritize customer segments for further investigation, and support business decision-making**, rather than to establish causal relationships or predict individual customer behavior with certainty.

8. Business Recommendations:

Based on the findings of the analysis, the following recommendations are proposed to support customer retention and improve customer engagement.

8.1 Prioritize the High-Risk Germany Customer Segment

The analysis identified a potentially high-risk segment consisting of customers in Germany with a Relationship Score of 0 and age above 40. This group contained 345 customers and had an observed churn rate of 75.36%.

The bank should first investigate the possible factors associated with the high churn rate in this segment. Customer feedback and additional behavioral information could be collected to understand their concerns and needs. Based on the findings, the bank could develop targeted retention strategies, such as personalized offers, improved services, or relevant product benefits.

However, the bank should avoid assuming that age or any individual characteristic causes churn, as the current analysis identifies an association rather than causation.

8.2 Investigate High-Balance but Inactive Customers

The analysis identified 2,903 customers with balances above the overall average who were classified as inactive. This group had an observed churn rate of 31.48%.

The bank should investigate why customers with relatively high balances remain inactive. Customer feedback and additional behavioral information could help identify potential factors such as service concerns, limited use of banking services, or the possibility that customers maintain the account as a secondary account.

Based on the feedback, the bank could develop appropriate engagement and retention strategies.

This finding also highlights that a high account balance should not automatically be considered an indicator of strong customer engagement or loyalty. Both financial value and engagement should therefore be considered when identifying customers who may require attention**.**

8.3 Learn from Customers with Two Products

Customers with two products showed the lowest observed churn rate at 7.58%, compared with 27.71% among customers with one product.

The bank could collect feedback from both one-product and two-product customers to understand differences in their experiences, product usefulness, service satisfaction, and engagement.

The objective should be to identify what characteristics or experiences are associated with the relatively strong retention of two-product customers and determine whether these insights can help improve the experience of one-product customers.

However, the bank should not assume that simply increasing the number of products will reduce churn, particularly because customers with three and four products showed much higher observed churn rates**.**

8.4 Improve Engagement Among Inactive Customers

Inactive customers had an observed churn rate of 26.85%, compared with 14.27% among active customers.

The bank should investigate the reasons behind customer inactivity through customer feedback and additional behavioral analysis. Possible areas of investigation could include service-related concerns, lack of awareness or guidance, limited use of banking services, or whether the customer is using the account as a secondary account.

Based on the feedback and identified customer needs, the bank could develop targeted engagement and retention strategies.

The goal should not simply be to make every customer more active, but to understand what prevents meaningful engagement and address those factors where appropriate.

8.5 Overall Recommendation

Overall, the bank should adopt a data-driven and customer-feedback-based retention approach rather than applying the same strategy to all customers. The analysis can help identify which customer segments require attention, while customer feedback and additional behavioral information can help determine why these segments may be at greater risk**.**

9. Conclusion:

This project provided insights into the factors and customer segments associated with customer churn and retention. The analysis showed that customer engagement, product utilization, geography, and customer characteristics can be useful for identifying groups with different levels of observed churn.

One important finding was that a high account balance does not necessarily indicate high customer engagement. The identification of high-balance but inactive customers showed that financial value and engagement should not be treated as the same concept.

The analysis also highlighted the importance of regular customer feedback. Feedback can help the bank understand customer concerns and identify service or engagement gaps that cannot be explained using the available dataset alone. Such information can then support more targeted retention strategies.

Another important learning from the project is the value of identifying and retaining engaged customers. Customers who demonstrate stronger engagement and product relationships showed lower observed churn in this dataset. Therefore, the bank should not focus only on customers who are already at high risk; it should also understand what helps engaged customers remain connected to the bank and use those insights to strengthen long-term relationships.

Finally, the project demonstrates that customer retention and customer acquisition should be considered together. While attracting new customers is important for business growth, retaining existing customers and building strong long-term relationships can also contribute significantly to sustainable business performance.

The findings should be interpreted as associations observed within the dataset rather than evidence of causation. The analysis identifies customer segments and characteristics associated with higher or lower observed churn, but additional data and statistical validation would be required to determine the underlying causes of customer churn.

Overall, the project demonstrates how data analysis can help a bank move from simply measuring churn to identifying priority customer segments, investigating potential risk factors, and developing more informed retention strategies.

10. References:

1. Unified Mentor. Internship project resources and dataset provided as part of the internship program.
2. Python Software Foundation. *Python Documentation.*
3. Pandas Development Team. *Pandas Documentation.*
4. Streamlit. *Streamlit Documentation.*